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Problem-Solving Style and Fit in Consulting and Auditing

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ABSTRACT: This research examined the problem-solving styles of public accountants in consulting and audit functions. Consultants were, on average, significantly more innovative in their problem-solving style than auditors. A path analysis indicated that accounting professionals whose preferred problem-solving style did not fit the demands of the job experienced greater role stress. The additional stress negatively affected job satisfaction, organizational commitment, and intent to stay with the firm. The results imply that matching an individual's problem-solving style to his or her functional role may help to minimize role stress and its attendant dysfunctional effects in public accounting.

Key Words: Adaption-Innovation theory, Problem-solving style, Role stress.

Data Availability: Contact the first author.

Public accounting organizations are generally partitioned into functional areas, two of which are audit or assurance services and consulting services. Audit services are performed in accordance with Generally Accepted Auditing Standards (GAAS), and auditors attest as to whether financial statements are prepared under Generally Accepted Accounting Principles (GAAP). These standards endow structure to the audit function in that they provide a framework for problem solving and limit the range of potential solutions. In contrast, the consulting function is neither standard-intensive nor guided by established rules. Consulting can be characterized as innovative-intensive in that consultants perform their services in a less-structured environment than do audit professionals and are often called upon to solve novel problems requiring creative and unique solutions (AAA 1986; Elliott 1992; Hood and Koberg 1991).

It has been well over a decade since the Bedford Committee (AAA 1986, 173) recognized that nontraditional accounting services, such as consulting, were more innovative-intensive than standard-intensive. More recently, the AICPA, in its Vision Statement (AICPA 1998), emphasized the increasing importance of consulting services and a shift from a standards-focused profession to one driven by opportunities resulting from rapidly changing technologies and markets. To the extent that the consulting function in public accounting operates in a less-structured environment and is often challenged by a more diverse set of problems than the audit function, it is expected that the cognitive problem-solving abilities demanded of consultants will differ from those their audit counterparts (AICPA 1998; Rahman and Zanzi 1995).

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This study employs a well-established measure from organizational psychology, the Kirton Adaption-Innovation Inventory (Kirton 1987), to examine how the fit between job demands and problem-solving style moderates role stress in affecting the outcome variables of job satisfaction, organizational commitment, and turnover intentions. Based upon a sample of audit and consulting professionals from a national public accounting firm, we find that consultants are significantly more innovative in their problem-solving style than are audit professionals. Importantly, we find that the congruence of problem-solving style with the occupational demands of the consulting and audit functions has a direct influence on role stress, which in turn impacts job satisfaction, organizational commitment, and intent to stay with the firm.

This study has implications for both the public accounting profession and accounting educators. First, the results indicate the importance of matching an individual's preferred problem-solving style to the demands of the job task in reducing role stress and its attendant dysfunctional effects in the domains of consulting and auditing. Second, by providing an appropriate fit between an individual's problem-solving style and the requirements of the occupational role, public accounting firms may be able to reduce the profession's reported 20 to 25 percent annual turnover (Shellenbarger 1998). Finally, prior research (Arunachalam et al. 1997; Gul 1986; Sweeney et al. 1999; Wolk and Cates 1994) indicates that students who self-select into undergraduate accounting programs are, on average, less innovative than the general population. This may result in a mismatch between the needs of the public accounting profession, which is demanding more innovative accountants (AAA 1986; AICPA 1998; *Perspectives* 1989; Hood and Koberg 1991) and the preferred problem-solving style of new entrants. This disparity provides accounting educators with the opportunity to create curricula more attractive to innovative students and responsive to the demands of the profession (Wolk et al. 1997).

In Section I, we discuss the theoretical underpinnings of this study. Hypotheses are developed in Section II. Section III contains our description of the measures and the sample employed in this research. The statistical analyses follow in Section IV, and we conclude the paper in Section V.

I. THEORETICAL DEVELOPMENT

Adaption-Innovation Theory

Public accounting firms compete in a complex world economy, increasingly driven by and dependent upon information technology. The proliferation of information technology has led to an expansion in both the size and importance of the consulting function in the public accounting industry and consultants are often called upon to resolve novel client problems requiring innovative solutions (AICPA 1998). Both individuals (Elliott 1992; Hood and Koberg 1991) and organizations (AAA 1986; AICPA 1998; *Perspectives* 1989) have recognized the need for public accountants who can operate effectively in this dynamic and unstructured environment.

An area of organizational psychology that is relevant to the cognitive problem-solving style of public accountants, especially in regard to the resolution of novel and unstructured problems, is Adaption-Innovation (A-I) theory (Kirton 1976, 1994). A-I theory relates to differences in the cognitive (thinking) style of individuals, with particular reference to the dimensions of creativity and problem solving. Although Huber (1983) criticized cognitive style research as a basis for MIS and DSS design, he notes that cognitive style research may be useful for career counseling, selection and placement, or coaching and training. The likely impact of the present research is in precisely these three areas.

Kirton (1994), who developed the theory as a result of his research on organizational change, contends that A-I is an appropriate framework for examining issues of creativity and problem solving as applied to occupational groups. A-I theory proposes that individuals exhibit, to a greater or lesser extent, one of two distinct problem-solving styles. The styles can be thought of as existing on a continuum, with extreme adaptors at one end and extreme innovators at the other. The adaptive style is characterized by a preference to do things better, while the innovative style is characterized by a preference to do things differently (Kirton 1988, 69). Although both adaptors and innovators are needed by organizations for their survival, the two styles reflect contrasting strengths and weaknesses. In an organizational context, a person will likely perform best when the demands of the job or occupation are congruent with his or her preferred problem-solving style (Foxall and Payne 1989; Kirton 1980, 1994). Table 1 illustrates the different approaches to problem solving taken by adaptors and innovators.

When faced with a problem, adaptors prefer the solution with the least impact on the assumptions, procedures, and values of the organization; they seek to improve the existing framework and use methods that are safe, secure, and predictable. Adaptors are characterized as being methodical, efficient, reliable, and sensitive to group cohesion and cooperation. Although they are essential for the day-to-day functioning of the organization, adaptors have difficulty moving out of established roles in times of change (Kirton 1976). Adaptors tend to produce fewer ideas than innovators, and their ideas or solutions are generally consistent with prevailing practices. An important focus of A-I theory is on the degree of structure preferred by an individual when problem solving or thinking in general. Adaptors favor well-established, structured situations and prefer being consistent with the prevailing reference system (Holland et al. 1991). Audit services, with its guiding framework of standards and an emphasis on compliance, represent the type of relatively structured occupations preferred by adaptors.

In contrast to their more adaptive counterparts, innovators prefer relatively unstructured environments and are at their best when challenged by unique problems requiring novel solutions. Innovators excel at producing new ideas, often cutting across existing paradigms, and are catalysts for organizational change. While innovators are essential in times of change, they have difficulty applying themselves to standardized tasks and ongoing organizational demands (Kirton 1976). Innovators can be insensitive to the traditional boundaries of the prevailing paradigm, challenge established methods and customs, but often bring about the radical changes "without which institutions tend to ossify" (Kirton 1976, 623). Consulting services, which lack the structure imposed by formal standards and often require new approaches in resolving problems, are expected to be especially attractive to more innovative public accountants. Kirton (1994, 69) contends that consultant groups are most effective when membership is dominated by innovators.

Goldsmith (1985) notes that much of the prior research on problem solving has had mixed results because no differentiation was made between level and style. An important distinction in A-I theory, however, is between level and style constructs. A-I theory is concerned with an individual's preferred problem-solving style as compared to his or her capacity (ability or intelligence) for problem solving (Foxall and Hackett 1992; Kirton 1994). Additionally, A-I theory is nonpejorative in that one style of problem solving is not seen as better or worse than the other (Kirton 1976, 1994). Adaptors and innovators differ, but either style, given a specific set of job demands, can be the more appropriate problem-solving approach. This is distinct from a level construct, such as intelligence or problem-solving ability, where higher ability is seen as better than lower ability.

Researchers have found that innovators are more likely to choose occupations characterized by less structure and more creativity, while adaptors gravitate toward relatively predictable and more-structured occupations (Kirton 1994). Foxall (1986), in a small sample study, found that cost accountants were, on average, significantly more adaptive than financial accountants. Gul (1986) found more adaptors than innovators in a sample of Australian accounting students and also found a relationship between problem-solving style and career preferences. Adaptors were interested in careers in auditing and banking, while innovators were interested in positions in management consulting. Sweeney et al. (1999) found that U.S. undergraduate accounting majors were significantly less innovative than the general population, with approximately 70 percent of their sample classified as adaptors. Wolk and Cates (1994) found that accounting students were significantly more adaptive in their problem-solving style than were other business majors. Arunachalam et al. (1997) found that accounting students classified as innovators performed significantly better on unstructured accounting tasks than students classified as adaptors.

TABLE 1
Problem-Solving Approaches of Adaptors and Innovators

Adaptor	Innovator
Employs disciplined, precise, methodical, efficient approach	Approaches tasks from unusual angles
Is concerned with resolving problems, rather than finding them	Discovers problems and discovers avenues of solution
Is an authority within structured situations	Takes control in unstructured situations
Seeks to resolve problems with tried and accepted means	Treats accepted means with little regard
Produces few ideas, generally aimed at improving existing system	Produces numerous ideas, some of which appear unsound, generally aimed at changing the existing system
Values continuity, stability, consensus, and group cohesion	Is catalyst to settled groups and irreverent of consensus, custom, and group norms
Is capable of doing routine work for long periods	Has little tolerance for routine and structure
Provides a safe base for the innovator's riskier operations	Provides the dynamics to bring about periodic radical change

Source: Adapted from Kirton (1976).

Research (Holland et al. 1991; Kirton 1980) indicates that managers may differ in their innovativeness across functional departments in the same organization. Managers in departments characterized by structure and routine were less innovative than managers in departments characterized by change and boundary spanning. Rahman and Zanzi (1995) found that consultants in public accounting had a preference for less organizational structure than auditors. This result is consistent with the notion that consultants are more likely to have an innovative problem-solving style (preference for low structure) and auditors are more likely to have an adaptive problem-solving style (preference for high structure).

Research indicates that a person's A-I problem-solving style is formed during childhood, changes little over time and is not readily alterable through training (Kirton 1987, 1994). The demands of a particular job or situation, however, may cause an incongruence between preferred problem-solving style and actual behavior. Coping behavior results when such circumstances force individuals to operate in a manner that differs from their preferred style (Clapp and de Ciantis 1989; Kirton 1994). While an individual may perform job tasks effectively by employing coping behavior, research has indicated that coping behavior results in increased job stress (Hayward and Everett 1983; Kirton and McCarthy 1988; Lindsay 1985; Thomson 1980). Kirton (1994, 30) notes that an "excessive demand for coping-behavior will not cause a change in cognitive style" but will cause stress and dissatisfaction "giv[ing] rise to a desire to leave the situation." An innovator (adaptor) employed in a position requiring primarily adaptive (innovative) problem-solving capabilities will feel additional stress as he or she attempts to cope with the situation. Alternatively, stress will be reduced when the demands of the position match or fit the person's preferred problem-solving style. By matching individual problem-solving styles with the demands of the job, public accounting firms may be able to improve performance while reducing stress (Kirton 1994).

The person-environment fit approach suggested by Kirton (1994) has often been used by researchers to study stress (Haskins et al. 1990) and is consistent with the definition of Edwards (1988, 242): "Stress is a negative discrepancy between an individual's perceived state and desired state, provided that the presence of this discrepancy is considered important by the individual." As Haskins et al. (1990, 362) note, this conceptualization of stress "focuses on the relationship between an individual's preferences and his or her job-related opportunities, constraints, and demands." Thus, whether a particular role is stressful to the individual depends upon the fit between his or her preferences and the demands of the job.

Prior research has consistently portrayed public accounting as a stressful profession (Collins and Killough 1992; Haskins et al. 1990; Smith 1990; Weick 1983). Negative effects of excessive stress in public accounting include diminished job performance (Choo 1986), reduced job satisfaction (Collins and Killough 1992; Rebele and Michaels 1990; Senatra 1980; Snead and Harrell 1991), job tension (Collins and Killough 1992; Smith et al. 1993), an increased desire to leave the organization (Collins and Killough 1992) and actual turnover (Collins 1993).[1] Excessive stress has also been found to impair the psychological and physiological well being of public accountants (Smith et al. 1993).

II. HYPOTHESES

The prior discussion indicated that the problem-solving style demanded of public accountants employed in more innovative-intensive services, such as in consulting, will differ from public accountants employed in more structured or standards-intensive services, such as in auditing. Based upon the literature review, we posit that consultants in public accounting will have a preference for an innovative problem-solving style and that auditors will have a preference for an adaptive problem-solving style.

> **H1:** Public accountants employed in consulting functions will be more innovative than public accountants employed in audit functions.

Although our expectation is that consultants are more likely to be innovators and audit professionals are more likely to be adaptors, each problem-solving style will be represented to a greater or lesser degree in both functional areas. When an individual is in an occupational role emphasizing a problem-solving style that conflicts with his or her own preference (i.e., low problem-solving style fit), coping behavior will result (Clapp and de Ciantis 1989). While coping behavior temporarily bridges the divide between one's preferred problem-solving style and the approach demanded under the circumstances, the cost to the individual is increased role stress and a desire to leave the situation (Hayward and Everett 1983; Kirton 1994; Lindsay 1985).

> **H2:** Consultants and auditors in public accounting whose cognitive problem-solving style fits the approach demanded by their functional area will experience less role stress than consultants and auditors whose problem-solving style does not fit the approach demanded by their functional area.

> **H3:** Consultants and auditors in public accounting whose cognitive problem-solving style fits the approach demanded by their functional area will have a greater desire to stay with the organization (lower turnover intentions) than consultants and auditors whose problem-solving style does not fit the approach demanded by their functional area.

Prior research has found that the increased role stress experienced by the individual due to lack of fit between his or her preferred problem-solving style and the demands of the occupational role will result in lower job satisfaction (Hayward and Everett 1983; Kirton 1994; Lindsay 1985). Our expectation is that the higher role stress experienced by adaptive problem solvers in the consulting area and by innovative problem solvers in the auditing area will result in lower job satisfaction.

> **H4:** Role stress will be negatively related to the job satisfaction of consultants and auditors.

Prior accounting research has consistently found a causal link between job satisfaction and organizational commitment in affecting turnover intentions (Aranya and Ferris 1984; Ferris and

[1] Although relatively low levels of stress have been linked to improved job performance in public accounting (Choo 1986), higher levels of stress, particularly over sustained periods, have consistently been associated with dysfunctional outcomes.

Aranya 1983; Lachman and Aranya 1986). Gregson (1992) investigated the causal ordering among the variables of job satisfaction, organizational commitment, and turnover intentions. He found the strongest model resulted when job satisfaction was antecedent to organizational commitment in predicting turnover intentions.

H5: Job satisfaction will be positively related to the organizational commitment of consultants and auditors.

Accounting researchers have found that both job satisfaction and organizational commitment are inversely related to turnover intentions (Bline et al. 1991; Gregson 1992). We also expect that an inverse relationship will result between the job satisfaction and turnover intentions and between the organizational commitment and turnover intentions of auditors and consultants.

H6: Job satisfaction will be positively related to the intentions of consultants and auditors to stay with the organization.

H7: Organizational commitment will be positively related to the intentions of consultants and auditors to stay with the organization.

III. METHOD

Measures

The Kirton Adaption-Innovation Inventory (KAI) was used to measure the adaptor-innovator cognitive problem-solving style construct (Kirton 1987). The KAI is a self-report instrument comprised of 32 five-point-scaled items, on which a respondent indicates the degree of ease or difficulty with which he or she could consistently maintain specified innovative and adaptive behaviors over time. KAI scores can potentially range from 32 to 160, but generally fall between 45 and 145. Extensive tests on general population samples indicate scores are normally distributed around a mean of 95.3 (Kirton 1994, 1987). Persons below the mean are referred to as adaptors and those above the mean as innovators.

In addition to computing a general score distinguishing adaptors and innovators, the KAI computes a score for each of three subscales. The first subscale measures sufficiency vs. proliferation of originality (SO). Adaptors prefer to produce fewer original ideas and tend to produce only the minimum number of problem solutions they perceive as useful and germane to the situation. Innovators prefer to proliferate ideas and tend to produce many solutions whether or not they are needed, or even plausible (Kirton 1976, 1994). The lower (higher) the SO score, the more adaptive (innovative) is the individual.

The second subscale measures efficiency (E). Adaptors prefer thoroughness, precision, reliability, and attention to detail. Innovators tend to ignore details, are less concerned with immediate efficiency and more concerned with long-term gains (Kirton 1976). The lower the E score, the more efficient (adaptive) is the individual.[2]

The third subscale measures rule or group conformity (R). Adaptors tend to be risk-averse and express a need for certainty, rules, and norms. As a result, they are more likely to conform to extant rules or to the prevailing paradigm. Innovators tend to be less risk-averse and less conforming to the rules, the group, or the existing paradigm. High R scores (more innovative) are indicative of a lack of conformity to groups and norms.

KAI scores do not correlate significantly with IQ, achievement tests, or level tests of creativity or cognitive complexity, but do significantly relate to other tests of style, such as resistance to change, risk taking, and intolerance of ambiguity (Kirton 1987, 1994). Foxall and Hackett (1992) confirmed that the KAI does not confound problem-solving ability with problem-solving style. Factor analysis (Foxall and Hackett 1992; Taylor 1989) has corroborated that the KAI loads on the

[2] Each component of the KAI is scored such that a low number is indicative of an adaptor. Thus, the E subscale may seem counter-intuitive: a low score indicates high efficiency and a high score indicates low efficiency.

three-factor structure (SO, E, and R). Researchers have found that KAI scores are not significantly correlated with measures of social desirability (Goldsmith and Matherly 1986, 1987; Elder and Johnson 1989). Kirton (1987) reports internal reliability scores (Cronbach's alpha) ranging in the high .80s for the KAI across multiple large samples and test-retest coefficients ranging from .82 to .91.

Role stress was measured by the "role conflict" scale developed by Rizzo et al. (1970). Role conflict (ROLECON) has been extensively used as a stress measure (Bamber et al. 1989; Collins and Killough 1992; Rasch and Harrell 1989, 1990; Rebele and Michaels 1990; Senatra 1980) and was used here to focus on the potential conflict between an individual's preferred problem-solving style and that demanded by the role.

We operationalized problem-solving style fit by formulating a variable reflecting the congruence/incongruence between an individual's preferred problem-solving style with that demanded by the applicable functional area. The problem-solving style fit variable (FIT) was based upon the Z-score of an individual's KAI score (KAIZ) and the percentage of time he or she spent in each functional area. The KAIZ score was computed as the subject's KAI score less the population mean of 95.3, divided by the population standard deviation of 17. This measure indicates the relative distance of a subject's KAI score from the mean. In theory, an individual with a negative KAIZ score (an adaptor) is best suited for structured, standard-intensive tasks, while an individual with a positive score (an innovator) is best suited for unstructured, innovative-intensive tasks. For an audit professional, a negative (positive) Z-score would indicate a better (worse) fit between problem-solving style and job demands. For a consulting professional, a positive (negative) Z-score would indicate a better (worse) fit between problem-solving style and job demands. To create an overall measure of fit between problem-solving style and job demands, the relationships above were modeled in the following equation:

$$\text{FIT} = (\text{KAIZ} * \%\text{ of consulting work}) - (\text{KAIZ} * \%\text{ of audit work}).$$

The interpretation of the resulting FIT variable is the higher (lower) the score, the better (worse) the fit of problem-solving style with job task. Positive FIT scores are reflective of a relatively better fit between an individual's preferred problem-solving style, as measured by the KAI, and the style demanded by functional area. Negative FIT scores indicate a relatively worse fit. The mean FIT score for the sample was 23, with a range of −225 to 219.

Outcome variables were measured with scales validated in prior research. Job satisfaction (JOBSAT) was measured using the three-item Hackman and Oldham (1975) scale. Organizational commitment (ORGCOM) was measured using the nine-item Mowday et al. (1979) scale. Turnover intentions were measured by the two-item, intent-to-stay scale (INTSTAY) from Dougherty and Pritchard (1985).

Sample

The management of a national public accounting firm agreed to allow firm personnel to participate in the study. Managing partners from 12 office locations in seven U.S. states provided data regarding the number of professionals in their respective offices. The appropriate number of self-contained packets, consisting of the KAI, a survey instrument incorporating the measures of interest and demographic questions, instructions, and a postage-paid return envelope, were sent to the office managing partners for distribution to accounting professionals. Respondents were not required to identify themselves and mailed the completed survey instrument directly to the researchers.

A total of 476 questionnaires where sent to the 12 participating offices. Respondents individually mailed their completed questionnaire back to the researchers. A total of 283 questionnaires were received for a 59 percent response rate. Three returned surveys were discarded because they were incomplete. In order to examine the sample for differences in cognitive problem-solving styles between consultants and their audit counterparts, it was necessary to classify individuals according to their functional area (audit or consulting). The question that each participant responded to was presented as follows:

Percentage of time spent in each functional area (total = 100%):

Audit____% Tax____% Consulting____% Other____%.[3]

For purposes of testing H1, if the respondent indicated that consulting (audit) activities occupied more time than any other category, then the respondent was classified as a consulting (audit) professional.[4] The respondents indicating that tax services occupied more time than any other category were deleted from the sample.[5]

The final sample consisted of 194 professionals composed of 51 consulting professionals and 143 audit professionals. The average age of the participants was 31.0 years, with 70 percent of the sample married, 35 percent female. Representation by level resulted in 34 percent staff, 32 percent seniors, 29 percent managers, and 5 percent partners.

IV. DATA ANALYSIS AND RESULTS

Initial analysis of the data in the form of means, standard deviations, and correlations is presented in Table 2. As indicated from the correlation matrix, subjects at higher position levels (POSIT) are more likely to be older, male, married, and working more hours. Subjects at more senior position levels have higher job satisfaction (JOBSAT), organizational commitment (ORGCOM), and intent to stay with the firm (INTSTAY). Position is also positively correlated with KAI scores, especially with regard to the R subscale, but not correlated with FIT. The FIT variable is correlated with ROLECON, but not with the outcome measures (JOBSAT, ORGCOM, INTSTAY) or POSIT.[6] Taken together, these results indicate that an individual's position within the firm is significantly related to demographic variables and outcome variables, but is not associated with the fit of an individual's problem-solving style and job demands. These relationships will be examined more fully in testing H2. As expected, the three outcome measures (JOBSAT, ORGCOM, INTSTAY) are highly correlated. Of final note is the high correlation between the outcome measures and the SO subscale. Individuals who tend to generate numerous ideas appear to hold more positive attitudes toward their public accounting work environment.

A comparison of consultants' and auditors' KAI scores, in addition to variables potentially relevant to working in a public accounting environment, is reported in Table 3. As hypothesized in H1, consultant's KAI scores (99.96) were significantly higher (more innovative: $F = 15.2$, $p < 0.001$) than the KAI scores of the audit professionals (90.91). A closer inspection of the subscales (E, R, and SO) of the KAI reveals that consultants had significantly higher scores on all three factors, with the largest difference attributable to the SO construct. For the E subscale, consultants scored 19.71, while auditors scored 17.65 ($F = 8.4$, $p < 0.004$). The auditor's lower score indicates a stronger preference for efficiency than do consultants. On the R subscale, consultants scored 34.90, while the auditors scored 32.86 ($F = 4.0$, $p < 0.047$). The auditor's lower score indicates a greater

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[3] The question was composed in this way because some accounting firms organize by industry area rather than by functional area. This allows individuals to disclose their primary functional roles regardless of their organizational structures.

[4] Two respondents indicated that equal amounts of time were spent in consulting and/or auditing and tax. These two were classified as consulting. Dropping or reclassifying these individuals did not affect the results.

[5] Tax services can be characterized as both standard-intensive and innovative-intensive. Tax professionals operate under the structure of revenue codes in performing compliance services, or alternatively, provide more innovative-intensive consulting services in the tax-planning role. Therefore, it is difficult to classify tax services as primarily standard-intensive or primarily innovative-intensive in that a tax accountant may provide both compliance services and tax-planning services. Thus, the 35 individuals indicating that taxation was their primary functional area were excluded from the analysis.

[6] The high correlation between FIT and KAI (including its subscales) is due to the computational relationship between the two measures. As the majority of the sample consisted of auditors, the significant negative correlation is due to disproportional sampling. In that the percentage of time worked in auditing is subtracted in the FIT equation, the negative correlation is a statistical artifact of the computation.

TABLE 2
Means, Standard Deviations, and Correlations

	Mean	Std.	Correlations													
	n = 194	Dev.	CONSULT	AGE	SEX	MAR	POSIT	HRS	FIT	KAI	E	SO	R	ROLECON	JOBSAT	ORGCOM
CONSULT	0.27	0.44	1													
AGE	31.01	9.70	0.28**	1												
SEX	1.35	0.48	−0.09	−0.14*	1											
MAR	1.70	0.46	0.07	0.22**	−0.27**	1										
POSIT	2.75	1.72	0.30**	0.63**	−0.17*	0.19**	1									
HRS	49.65	10.23	0.10	0.18*	−0.12	0.07	0.26**	1								
FIT	22.51	65.14	−0.02	−0.07	0.17*	−0.03	−0.05	−0.03	1							
KAI	93.31	14.87	0.27**	0.16*	−0.19**	0.02	0.21**	0.11	−0.41**	1						
E	18.20	4.48	0.20**	0.12	−0.15*	−0.01	0.23**	0.14*	−0.31**	0.70***	1					
SO	41.76	7.62	0.28**	0.08	−0.19**	0.08	0.07	0.10	−0.31**	0.82***	0.30**	1				
R	33.40	6.36	0.14*	0.19*	−0.11	−0.05	0.26**	0.03	−0.36**	0.85***	0.55**	0.49**	1			
ROLECON	4.05	1.03	−0.03	0.05	−0.11	0.02	0.09	0.14	−0.14*	0.21**	0.15*	0.11	0.25**	1		
JOBSAT	5.19	1.28	0.21**	0.19**	−0.08	0.10	0.23**	0.12	−0.04	0.12	−0.03	0.24**	0.02	−0.38***	1	
ORGCOM	4.91	0.96	0.19**	0.20**	−0.06	0.18*	0.19**	0.14	0.00	0.08	−0.06	0.24**	−0.05	−0.33***	0.67***	1
INTSTAY	3.51	1.29	0.29**	0.26**	−0.23**	0.22**	0.41**	0.08	−0.03	0.12	0.03	0.19**	0.05	−0.23**	0.65***	0.66***

*, **, *** Significant at 0.05, 0.01, and 0.001, respectively.

TABLE 3
Tests of Differences between Consultants and Auditors

	Audit (n = 143)		Consulting (n = 51)				
	Mean	**Std. Dev.**	**Mean**	**Std. Dev.**	**Difference**	**F**	**Significance**
AGE	29.37	8.73	35.54	10.86	−6.17	16.647	0.001
SEX	1.37	0.48	1.27	0.45	0.10	1.533	0.217
MAR	1.31	0.47	1.24	0.43	0.07	0.988	0.322
POSIT	2.44	1.65	3.61	1.64	−1.17	18.935	0.001
HRS	49.06	10.81	51.29	8.30	−2.23	1.828	0.178
FIT	27.60	71.82	13.60	72.00	14.00	0.062	0.804
KAI	90.91	13.90	99.96	15.57	−9.05	15.182	0.001
E	17.65	4.42	19.71	4.34	−2.06	8.361	0.004
SO	40.47	7.33	45.35	7.31	−4.88	16.965	0.001
R	32.86	6.15	34.90	6.74	−2.04	4.003	0.047
ROLECON	4.07	0.96	4.01	1.21	0.06	0.125	0.724
JOBSAT	5.03	1.33	5.63	1.03	−0.60	8.801	0.003
ORGCOM	4.81	0.97	5.21	0.90	−0.40	6.917	0.009
INTSTAY	3.29	1.30	4.11	0.90	−0.82	17.477	0.001

preference for operating within extant rules or norms. The consultants in the sample had a significantly greater preference ($F = 17.0$, $p < 0.001$) for producing original and creative ideas than did the audit professionals, with an average SO score of 45.35 compared to 40.47 for the audit group.

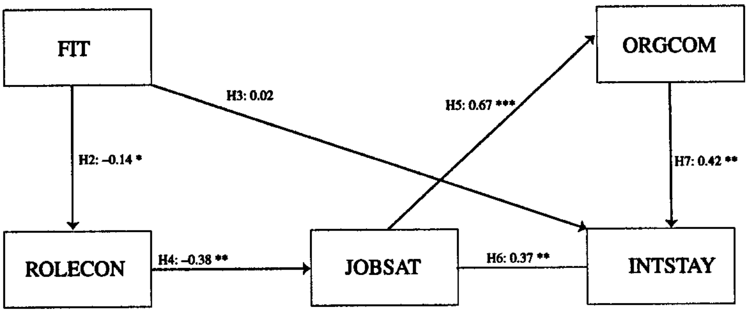
Tests of differences (Table 3) indicate that consultants had significantly higher job satisfaction ($F = 8.8$, $p < 0.003$) and organizational commitment ($F = 6.9$, $p < 0.009$) than auditors and reported a stronger intent to stay with the firm ($F = 17.5$, $p < 0.001$). This may be attributable to consultants being, on average, older by approximately six years, and more likely to hold a more senior position than the auditing participants.

A supplemental analysis was conducted to determine if the dichotomous classification of individuals into either consulting or auditing was driving the results reported in Table 3. This analysis involved regressing the continuous variable percent of TIME CONSULTING on each independent variable. Using the INTSTAY variable as an example, we regressed the percent of TIME CONSULTING on INTSTAY. The results revealed a significant and positive beta ($p < 0.001$), indicating that intent to stay with the firm increased with the percentage of time spent consulting. This parallels the result reported in Table 3, where individuals classified as consultants reported a higher INTSTAY score. The overall results of the supplemental analysis were consistent with the dichotomous classification of subjects as either auditors or consultants.[7]

At this point, the analysis moves from comparisons between consultants and auditors to testing the influence of participants' job fit on role stress and its subsequent effect on job outcomes, as posited in hypotheses H2 through H7. The theoretical model is presented in Figure 1. The analyses were conducted with the AMOS modeling software (SPSS 1997). All analyses used the maximum likelihood method parameter estimate and were performed on the raw data rather than on the variance/covariance matrix.

[7] A related issue involves the possibility that ROLECON could be influenced by the degree to which an individual spreads his or her time across functional areas. In other words, an individual who works exclusively in one area might have lower ROLECON than an individual who spends time in more than one area. We performed a sensitivity analysis to determine if ROLECON was influenced by the percentage of time worked in the individual's dominant area. The analysis revealed that ROLECON was not sensitive to time spread across functional areas.

FIGURE 1
Standardized Path Coefficients for the Impact of Problem-Solving Style Fit on Role Conflict, Satisfaction, Commitment, and Intent to Stay



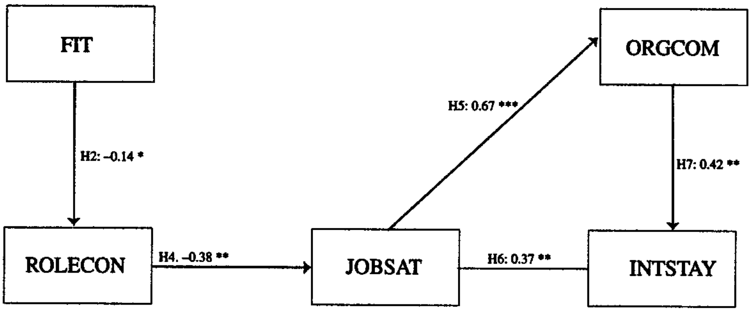
*, **, *** Significant at 0.05, 0.01, and 0.001, respectively.
Model statistics: chi-square = 10.51; df = 8; Chi-square/df = 1.31; significance = 0.23
Fit Indices: comparative fit index (CFI) = 1; normed fit index (NFI) = 0.98; incremental fit index (IFI) = 1; non-normed fit index (NNFI) = 0.99; Root Mean Square Error of Approximation (RMSEA) = 0.03

Several goodness-of-fit indices for the model are presented in Figure 1. The Chi-square statistic included in this figure provides a test of the null hypothesis that the covariance matrix has the specified model structure, i.e., that the model fits the data. An insignificant Chi-square indicates a good fit. Other measures of goodness-of-fit include: the NFI, the NNFI, the CFI, the IFI, and the RMSEA. The NFI may range in value from 0 to 1, where 0 represents the goodness-of-fit associated with the null model (one specifying that all variables are uncorrelated), and 1 represents the goodness-of-fit associated with a saturated model (a model with 0 degrees of freedom that perfectly reproduces the original covariance matrix). The NNFI and CFI are variations of the NFI that have been shown to be less biased in small samples (Bentler 1989). They also range in value from 0 to 1, where 1 represents a fit that is as good as the saturated model. Values on the NFI, NNFI, CFI, and IFI over 0.9 indicate an acceptable fit between the model and data. The RMSEA (Root Mean Square Error of Approximation) probability indicates overall model fit per degree of freedom. Browne and Cudeck (as reported in Bollen and Long 1993, 144) recommend that an absolute RMSEA value of less that 0.05 indicates a close fit and less than 0.08 suggests a reasonable fit.

Estimation of the theoretical model in Figure 1 revealed a nonsignificant model Chi-square value ($10.51_{8,194}$, $p < 0.23$). Values of NFI, NNFI, CFI, and IFI all exceeded 0.9 and the RMSEA was only 0.03, indicating an excellent fit between the model and data. The t-values for all path coefficients were statistically significant except for the path from FIT to INTSTAY. With the exception of this path, all the standardized path coefficients exceeded 0.14 in absolute magnitude, indicating that they were meaningful in size. Given the apparent small contribution of the path from FIT to INTSTAY, we evaluated a constrained model in which this path was not present. The goodness-of-fit indices for this constrained model are presented along with the model in Figure 2.

By comparing the Chi-square statistic for the initial model to the Chi-square statistic for the constrained model, it was possible to perform a Chi-square difference test to determine whether the deletion of the FIT to INTSTAY path resulted in a significant improvement in the fit of the model. This difference test was computed as $10.51 - 10.72 = -0.21$. With 2 df, the Chi-square difference statistic was insignificant, indicating that the constrained model did not perform better than the theoretical model. Removing the path from FIT to INTSTAY did not significantly improve the fit of the model. The implication is that both models fit well and that either model is equally acceptable. The more parsimonious of the two models (the constrained model) is chosen as the final model.

FIGURE 2
Standardized Path Coefficients for the Final Model of Impact of Problem-Solving Style Fit on Role Conflict, Satisfaction, Commitment, and Intent to Stay



*, **, *** Significant at 0.05, 0.01, and 0.001, respectively.
Model statistics: chi-square = 10.72; df = 10; Chi-square/df = 1.07; significance = 0.38
Fit Indices: comparative fit index (CFI) = 1; normed fit index (NFI) = 0.98; incremental fit index (IFI) = 1; non-normed fit index (NNFI) = 1; Root Mean Square Error of Approximation (RMSEA) = 0.01

All of the standardized path coefficients are meaningful in absolute magnitude, and t tests indicated that all were significant at $p < 0.05$. The analysis produced R^2 values of 0.15 for job satisfaction and 0.51 for intent to stay, indicating that 15 percent of the variance in job satisfaction and 51 percent of the variance in intent to stay are accounted for in the model.

FIT is hypothesized in H2 to negatively influence ROLECON. The negative path coefficient of 0.14 between FIT and ROLECON is significant ($p < 0.05$) and supports H2. Consultants whose preferred problem-solving style was more innovative (higher FIT scores) had lower role stress (ROLECON) than consultants whose preferred problem-solving style was more adaptive. Conversely, auditors who were more adaptive problem solvers (higher FIT scores) experienced lower stress (ROLECON) than did auditors who were more innovative problem solvers. As indicated in the discussion of the theoretical model in Figure 1, the path between FIT and INTSTAY hypothesized in H3 was insignificant and was therefore dropped from the final model.

As hypothesized in H4, a significant negative relationship ($p < 0.01$) existed between stress (ROLECON) and job satisfaction (JOBSAT). The paths between job satisfaction and organizational commitment (ORGCOM, $p < 0.001$) and between job satisfaction and turnover intentions (INTSTAY, $p < 0.01$) were also significant, as hypothesized in H5 and H6. Finally, as hypothesized in H7, the path between organizational commitment and turnover intentions was significant ($p < 0.01$).

The final model is substantially consistent with our hypotheses. Public accountants in auditing and consulting whose cognitive problem-solving style was (in)congruent with the approach demanded by the functional area experienced (more) less stress in the form of role conflict. As role conflict increased, job satisfaction decreased and negatively impacted organizational commitment and the individual's intent to stay with the firm.

V. CONCLUSION

This research examined the problem-solving styles of public accountants in consulting and auditing. As hypothesized, consultants attained significantly higher scores (more innovative) on the Kirton Adaption-Innovation Inventory than audit professionals. Consultants were found to be, on average, more cognitively inclined to produce creative and original ideas and function effectively in unstructured settings than their auditor counterparts.

The major contribution of this research is in the empirical examination of how the fit of an individual's preferred problem-solving style interacts with the demands of the consulting and auditing functions in affecting stress and outcome variables. The results are consistent with Kirton's (1994, 83) assertion that it is the individual's immediate occupational group within the organization that is the critical factor in determining if problem-solving style fit will occur. Consultants who preferred a more innovative problem-solving approach experienced less role stress and its attendant dysfunctional effects than did consultants who preferred a more adaptive problem-solving approach. Alternatively, auditors whose preferred problem-solving style was adaptive experienced less role stress than did auditors whose preferred problem-solving style was innovative.

These results have important implications for public accounting, which is becoming less structured, more systems-oriented, and increasingly reliant upon consulting services (AICPA 1998; Elliott 1992). By examining the relative amount of structure, creativity, and innovativeness required within each functional area, public accounting firms may be able to select individuals whose preferred problem-solving style matches the demands of the functional role. In this way firms may be able to reduce employees' role stress and positively affect their satisfaction, commitment, and intentions to stay with the organization.[8]

Accounting academicians and program administrators can play an important part in ensuring a supply of innovative accounting graduates sufficient to meet the demands of the profession. Researchers have suggested reducing the structure in accounting programs and adopting diverse instructional approaches designed to make the curriculum attractive to both innovators and adaptors (Sweeney et al. 1999; Wolk and Cates 1994). Accounting educators should also be aware that their own problem-solving style might influence their pedagogical approaches and propensity for implementing change (Wolk et al. 1997).

Interpretation of the results of this study is subject to some caveats, which may provide opportunities for future research. In examining the relationship between job fit and cognitive style, we focused only on problem-solving style, as operationalized in Adaption-Innovation Theory. Future researchers can expand knowledge of the job-fit dynamic by including other cognitive or personality factors in their design. Job fit is only measured indirectly and is dependent upon our classification of auditing as standard-intensive and consulting as innovation-intensive. As the sample was drawn from a single national public accounting firm, generalizability is limited to the extent to which the structure endowed in the audit and consulting functions of the sample firm is representative of the structure of those functions in the population of public accounting firms. The study also involved the analysis of self-reported data and may therefore be subject to problems inherent in cross-sectional data collection methodologies (Podsakoff and Organ 1986).

Future research can contribute to the extant body of literature by examining the impact of problem-solving style on the selection, socialization, and promotion of public accountants. Management of public accounting firms may also benefit from research directed toward understanding the relationship between individual problem-solving styles and performance in auditing and consulting.

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[8] A balance of problem-solving styles may be necessary within each job function. Prior research involving work groups and problem-solving style has generally shown that a mix of adaptive and innovative styles can bring the strengths (and weaknesses) of both styles to bear upon a situation.

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